MPC-PiNNs technique applied to Quadrotors

Elective in Robotics [UAV] Course Project

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2 Models

Unicycle

$$\begin{cases} \dot{x} = v\cos(\theta) \\ \dot{y} = v\sin(\theta) \\ \dot{\theta} = w \end{cases} \tag{1}$$

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4 Physical Informed Neural Networks

Physical Informed Neural Networks (PiNNs) propose an alternative approach where the solution is approxi- mated by a neural network that is trained not only on data but also by enforcing the underlying physical laws via the differential equations.

$$\mathcal{L}_{\text{data}}(\theta) = \frac{1}{N_u} \sum_{i=1}^{N_u} \left| u_\theta(\mathbf{x}_i, t_i) - u_i^{\text{obs}} \right|^2 \tag{2}$$

$$\mathcal{L}_{\text{phys}}(\theta) = \frac{1}{N_r} \sum_{j=1}^{N_r} \left| \mathcal{N}[u_\theta](\mathbf{x}_j, t_j) \right|^2 , \tag{3}$$

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5 Results

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6 Conclusions